

BUSINESS OVERVIEW

ASAR Sanctuary

An intelligent Quran reading companion built for the Quran Foundation Hackathon. Combines behavioral science, AI-powered reflection, and beautiful design to foster lasting daily Quran engagement through a 60-second micro habit architecture.

Live Demo: fnwholesale.pk
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1. Executive Summary

ASAR Sanctuary is a full-stack Quran reading and reflection companion that combines a modern React SPA with a robust FastAPI backend. The name ASAR reflects the app's mission: leaving a lasting trace (Arabic: asar) in your daily Quran reading journey. Whether a beginner setting a 5-minute daily goal or a seasoned reader completing the entire Mushaf, ASAR adapts to your pace and keeps you consistent.

The platform provides an immersive, distraction-free reading experience enhanced by AI-powered verse exploration, daily habit tracking with visual streak constellations, cloud-synced bookmarks, and a verified RAG-based chat companion, all grounded strictly in authentic Quranic text. The AI layer leverages LongCat, an OpenAI-compatible LLM service, to power two distinct chat pipelines: a Verified Assistant that uses strict anti-hallucination prompts with Quran API facts, and a REST RAG pipeline with multi-stage retrieval including query bridging, JSON query planning, relevance gating, neighbor context expansion, and optional alignment checking. Every AI-generated response is verifiably grounded in authentic sources.

The product addresses the fundamental problem that most Muslims want to read the Quran daily but life gets in the way. The number one reason people abandon Quran reading is not lack of interest but friction. Traditional apps present the entire 604-page Mushaf and say "go read," which is overwhelming. ASAR flips this model with a 60-Second Micro Habit Architecture, a behavior-science-driven system that makes daily Quran engagement so small and effortless that skipping it feels harder than doing it. The application is built as a Progressive Web App (PWA), enabling cross-platform installation on Android, iOS, Windows, and Linux devices, and is developed for the Quran Foundation Hackathon with a live demo at fnwholesale.pk.

2. The Problem: Why People Abandon Quran Reading

Most Muslims genuinely want to read the Quran daily, yet life consistently intervenes and the discipline required to maintain this sacred practice often falls short. The intention is present, but the execution remains elusive for the vast majority of believers around the world. Surveys and behavioral studies consistently show that while over 90% of Muslims consider daily Quran reading important, fewer than 20% actually maintain the practice beyond the first month of any serious attempt.

The number one reason is friction, not lack of interest. Traditional Quran applications present the entire 604-page Mushaf and essentially tell the user to "go read," which is overwhelmingly intimidating for most people, especially those beginning their spiritual journey. Research in behavioral psychology, particularly BJ Fogg's Tiny Habits framework and James Clear's Atomic Habits methodology, demonstrates that habit formation depends critically on starting small. When the required effort exceeds a person's motivational threshold, the habit never takes root.

A 60-second commitment bypasses the brain's resistance to large tasks entirely. By reducing the barrier to entry to a single ayah readable in under a minute, the friction drops to near-zero. ASAR's core trick is to lower the floor so much that everyone can step in, then let the architecture naturally raise the ceiling. Users who start with one verse per day typically find themselves spending 10 to 15 minutes exploring the Quran Reader, engaging with the AI Companion, and deepening their understanding, all without ever feeling pressured or overwhelmed by the scale of the text.

3. The 60-Second Micro Habit Architecture

ASAR Sanctuary implements a scientifically grounded three-pillar engagement system designed to sustain daily Quran reading through behavioral science principles. Each pillar addresses a different dimension of habit formation: visual reward, intellectual value, and proactive encouragement. Together, they create a self-reinforcing loop that transforms Quran reading from an aspirational goal into an effortless daily routine.

3.1 Three-Pillar Engagement System

Pillar 1 - Visual Momentum: The Streak Constellation and Flow Ring provide immediate, tangible gratification for marking an ayah as complete. The constellation grows with each consecutive reading day, creating a visual pattern that the human brain instinctively wants to continue. The Flow Ring fills progressively throughout the day, combining streak consistency with daily effort marks into a single, motivating visual indicator. Users can see their progress at a glance, turning abstract consistency into concrete visual rewards that reinforce the habit loop.

Pillar 2 - AI Reflection: The Chat Companion and AI Insights provide deep, personalized value during each reading session. The AI Companion answers questions grounded strictly in verified Quranic text, preventing hallucinated or inaccurate content. The Insights Engine generates personalized spiritual reflections based on the user's reading history and session patterns, giving users

a meaningful, intellectually enriching reason to return each day. This pillar transforms passive reading into an interactive learning experience.

Pillar 3 - Proactive Reminders: The Motivational Notification System triggers if a user visits the sanctuary but has not yet completed their daily goal. Using the "1% > 0%" philosophy, the system encourages action without being intrusive. This gentle nudge leverages loss aversion and the desire for completion to convert visitors into active readers, ensuring that even on low-motivation days, the habit loop stays intact.

3.2 Core Design Principles

Principle	Implementation in ASAR
Make it Tiny	One single ayah on the dashboard - readable in under 60 seconds, zero barrier to entry.
Make it Visual	Streak Constellation and Flow Ring turn consistency into tangible visual rewards that grow daily.
Make it Immediate	One-tap mark-complete - no navigation, no menus, no friction. The entire loop takes under a minute.
Make it Progressive	Reading cursor auto-advances - silently compounds into completed Surahs over weeks and months.
Make it Identity-Driven	Habits Heatmap reflects "You are someone who reads Quran every day" through visible patterns.
Make it Guided	Onboarding captures goal, time budget, topic interest; recommended verse engine delivers personally relevant ayahs.

Table 1: Core design principles of the ASAR Micro Habit Architecture

3.3 Flow Ring Details

The Flow Ring is the centerpiece visual element of the ASAR dashboard, combining two distinct components into a single progress indicator that motivates and guides the user throughout each day. The first component is the **Streak Base**, which contributes up to 55% of the ring based on the user's consecutive reading streak. A user on a 7-day streak, for example, would have approximately 55% of the ring already filled before reading a single verse, providing a strong motivational foundation.

The second component is the **Daily Effort**, which contributes 14% per ayah marked as complete during the current day. This means that users on a 1-day streak can still achieve a 100% Flow Ring by reading approximately 6 verses in a single session, ensuring that even new users feel a sense of accomplishment and

completion. After reaching 100%, Engagement Points continue accumulating, rewarding enthusiastic readers who go beyond their minimum goal. This dual-component design ensures that both new users and long-term readers find the Flow Ring motivating and achievable every single day.

4. Key Features Overview

ASAR Sanctuary offers a comprehensive suite of twelve core features designed to address every aspect of the Quran reading journey, from initial engagement to long-term habit formation. Each feature has been carefully designed to serve both the user's spiritual needs and the platform's business objectives of retention, engagement, and user satisfaction. The following table summarizes each feature along with its detailed description.

Feature	Description
Daily Ayah Dashboard	Personalized verse recommendation with audio recitation, one-tap mark-complete, and progress tracking. Surfaces a single ayah readable in under 60 seconds.
Streak Constellation and Flow Ring	Visual habit-building with a growing star-like constellation map and a dual-component progress ring combining streak consistency and daily effort marks.
Quran Reader and Focus Mode	Browse all 114 Surahs with distraction-free mode showing Arabic text, translation, and transliteration for deep spiritual reflection.
AI Companion (Verified RAG)	Questions answered grounded strictly in verified verse text through a multi-stage RAG pipeline - no hallucinated content, ever.
Habits Heatmap	Calendar-style heatmap of reading consistency over weeks and months, making patterns and gaps visible for self-reflection.
Insights Engine (AI-Powered)	Personalized spiritual reflections generated by LLM based on session history, reading patterns, and verse engagement data.
Premium Translation Selector	Searchable dropdown with grouped editions supporting 100+ languages for rapid discovery and language-switching.
PWA Excellence	Full Progressive Web App with guided cross-platform installation for Android, Desktop, and step-by-step iOS guidance.
Dhikr Counter	Persistent local Tasbeeh counter for tracking dhikr and spiritual remembrance practices across sessions.
Bookmark Library	Save verses with optional notes, cloud-synced to Quran Foundation account for cross-device access and continuity.

Quran.com OAuth	Secure sign-in using Quran Foundation OAuth2 with PKCE flow, enabling activity and bookmark synchronization.
Demo Mode	Try every feature instantly without linking an external account, eliminating onboarding drop-off completely.

Table 2: Core features of ASAR Sanctuary

5. Technology Stack

ASAR Sanctuary is built on a modern, production-ready technology stack that prioritizes developer productivity, type safety, and performance. The backend leverages Python's async ecosystem with FastAPI for high-throughput API handling, while the frontend uses React 18 with TypeScript for type-safe component development. The following tables detail the complete technology stack organized by backend and frontend categories.

5.1 Backend - ASAR Engine

Technology	Purpose
FastAPI	Async API framework with automatic OpenAPI documentation at /docs endpoint
SQLAlchemy	ORM for SQLite persistence with domain models and schema validation
Pydantic	Settings management, request/response schemas, and environment variable validation
httpx	Async HTTP client for Quran API calls with connection pooling and timeout handling
AsyncOpenAI	OpenAI-compatible client for LongCat LLM completions powering chat pipelines
python-jose	JWT encoding, decoding, and signature validation for authentication tokens
passlib	Password hashing for user credentials and demo account management
pytest	Backend testing with pytest-asyncio, httpx, and anyio for comprehensive async test coverage

Table 3: Backend technology stack - ASAR Engine

5.2 Frontend - Sanctuary SPA

Technology	Purpose
React 18	UI library with hooks and context for state management across components
TypeScript	Type-safe development with full type coverage across all components and utilities
Vite	Fast build tool and dev server with hot module replacement for rapid iteration
React Router	Client-side routing with authentication guards and onboarding gates
Axios	HTTP client with session ID and Bearer token interceptors for API communication
Vitest	Unit testing framework for frontend components, hooks, and utility functions
Playwright	End-to-end testing for critical user workflows across multiple browsers

Table 4: Frontend technology stack - Sanctuary SPA

6. Quran Foundation Hackathon Compliance

ASAR Sanctuary fulfills all hackathon requirements by integrating both mandated API categories with multiple endpoints from each. The Content API provides the foundational Quranic data layer, while the User API enables identity management and cross-platform data synchronization. This comprehensive integration ensures that ASAR delivers genuine value built on the Quran Foundation's infrastructure, not merely superficial API connectivity.

6.1 Content API Category

ASAR utilizes multiple Content API endpoints to deliver a rich Quranic reading experience. The **Quran APIs** provide core verse retrieval and surah metadata, forming the backbone of the reading experience. The **Audio APIs** supply integrated recitations for every ayah, enabling users to listen while reading. The **Tafsir APIs** provide verified grounding for the AI Companion, ensuring all AI responses are anchored in authentic scholarly commentary. The **Translation APIs** support over 100 languages through the Premium Translation Selector, making the platform accessible to a global audience regardless of their primary language.

6.2 User API Category

On the User API side, ASAR implements full **bi-directional bookmark synchronization**, allowing users to save verses with notes on any device and have those bookmarks instantly available everywhere. The **Streak Tracking integration** syncs reading activity data in real-time to the Activity Days endpoint, ensuring that a user's reading streak persists even when switching between devices or platforms. The **Identity** layer uses secure authentication via Quran Foundation OAuth2 with PKCE flow, providing enterprise-grade security while maintaining a seamless user experience that requires only a single click to authorize.

API	Category	Integration Details
Quran APIs	Content	Core verse retrieval and surah metadata for the entire Quran
Audio APIs	Content	Integrated recitations for every ayah with multiple reciter options
Tafsir APIs	Content	Verified grounding for AI Companion to prevent hallucinated responses
Translation APIs	Content	Support for 100+ languages through the Premium Translation Selector
Bookmarks	User	Full bi-directional cloud synchronization across all linked devices
Streak Tracking	User	Real-time sync to Activity Days endpoint for persistent reading streaks
Identity (OAuth2)	User	Secure authentication via Quran Foundation OAuth2 with PKCE flow

Table 5: Quran Foundation Hackathon API compliance matrix

7. Getting Started

Setting up ASAR Sanctuary for local development requires a few prerequisite tools and services. The platform is designed for rapid onboarding so that developers can begin contributing within minutes. This section outlines the complete setup process from initial requirements to running the live application on your local machine.

7.1 Prerequisites

Before beginning the setup process, ensure that your development environment meets the following minimum requirements: **Python 3.11+** is required for the FastAPI backend, as the codebase leverages modern Python features including async/await patterns, type hints, and match statements. **Node.js 18+** is required

for the React frontend and its associated toolchain including Vite, TypeScript, and npm packages. A **LongCat API key** is required for the AI chat pipelines, as all LLM completions are routed through this OpenAI-compatible service. Finally, **Quran Foundation OAuth credentials** are required for enabling the full authentication flow with bookmark and streak synchronization.

7.2 Setup Process

The setup process begins by cloning the repository from GitHub. For the backend, navigate to the Back-End directory, create a Python virtual environment, install dependencies with pip, and configure the .env file with your API keys and OAuth credentials. For the frontend, navigate to the Front-End directory, install npm dependencies, and configure the frontend .env file with the backend API URL. The backend runs on port 8000 and the frontend development server runs on port 5173, with Vite proxying API requests to the backend automatically during development.

7.3 Live Demo

A fully functional live demo is available at <https://fnwholesale.pk/>, allowing anyone to explore every feature of ASAR Sanctuary without setting up a local development environment. The demo includes the Demo Mode feature, which provides instant access to all functionality without requiring account creation. This makes it easy for hackathon judges, potential users, and contributors to experience the platform's capabilities immediately.

8. Product Roadmap and Future Vision

ASAR Sanctuary is positioned as a platform that can expand significantly beyond its hackathon origins. The 60-Second Micro Habit Architecture is a universally applicable behavioral framework that could extend to any daily spiritual practice, from dhikr and dua to hadith study and Islamic lecture series. The modular architecture (FastAPI backend + React SPA) was intentionally designed for rapid iteration and feature development, making the platform well-suited for continued growth and community-driven evolution.

Several exciting future directions have been identified for the platform's continued development. **Multi-language support expansion** would make the interface itself available in Arabic, Urdu, Malay, Turkish, and other high-demand languages, complementing the existing 100+ language translation support for Quranic text. **Community features and group reading challenges** would introduce a social dimension to the platform, allowing friends, families, and

mosque communities to read together and encourage one another through shared streaks and collaborative goals.

Advanced AI insights with deeper tafsir integration would enhance the spiritual depth of the AI Companion by incorporating classical tafsir sources alongside modern scholarly commentary. **Offline-first capabilities** would serve users in areas with limited or unreliable internet connectivity, ensuring that the daily reading habit is never interrupted by technical barriers. Finally, **integration with additional Islamic content sources** such as hadith collections, supplication databases, and scholarly lecture libraries would transform ASAR from a Quran reading companion into a comprehensive Islamic knowledge and spiritual growth platform.

9. Support and Community

ASAR Sanctuary benefits from a growing community of users, contributors, and supporters who are passionate about leveraging technology to enhance daily Quran engagement. Multiple channels are available for support, feedback, and collaboration, ensuring that every user and developer can find the help they need to get the most out of the platform.

For direct support, the platform provides an integrated email channel at **Hackathon@quran.com** for technical questions, bug reports, and feature requests. The **Discord Community** at <https://discord.gg/SpEeJ5bWEQ> serves as the primary hub for real-time discussion, community events, and collaborative development. The complete source code is open-source and available on **GitHub** at <https://github.com/MahrZain/Quran-Hackathon.git>, licensed under the permissive **MIT License**, which encourages both personal and commercial use, modification, and distribution.

The MIT License ensures that ASAR Sanctuary remains freely accessible to everyone, from individual users seeking to enhance their personal Quran reading practice to organizations looking to build derivative applications on top of the platform's solid foundation. Community contributions are welcomed and encouraged, whether through code submissions, bug reports, feature suggestions, translations, or documentation improvements. The project's modular architecture makes it straightforward for developers to extend the platform with new features, integrations, and capabilities.